Biotechnology and genetic modification

20 questions: 10 multiple choice, 10 structured.

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1 · Part 1 Core Biotechnology and genetic modification [1]

Why are bacteria useful for producing large quantities of a useful protein?

A They have a nucleus containing many chromosomes.

B They can reproduce rapidly and make complex molecules.

C They can only make simple molecules such as glucose.

D They reproduce slowly, so cultures are easy to control.

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2 · Part 1 Core Biotechnology [1]

Which yeast product makes bread dough rise?

A lactic acid

B oxygen

C cellulose

D carbon dioxide

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3 · Part 1 Core Biotechnology [1]

Which product of anaerobic respiration in yeast is used as a biofuel?

A ethanol

B urea

C starch

D oxygen

4 · Part 1 Core Biotechnology [1]

Why is pectinase added during some fruit-juice production?

A to convert all juice into protein

B to sterilise the juice by boiling

C to break down pectin and help release juice

D to turn sugar into oxygen

5 · Part 1 Core Biotechnology [1]

Which comparison best tests the effect of enzymes in washing powder?

A different stains washed at different temperatures with different powders

B one fabric washed and another left completely dry

C different fabrics washed for different durations

D equal stained fabrics washed under identical conditions with enzyme and enzyme-free powders

6 · Part 1 Supplement Biotechnology [1]

Which enzyme is used to make lactose-free milk?

A pectinase

B lactase

C lipase

D amylase

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7 · Part 1 Supplement Biotechnology [1]

Why is cooling often needed in an aerobic industrial fermenter?

A Microorganisms absorb heat from the surroundings as they grow.

B Cooling removes waste products from the culture.

C Respiration by microorganisms releases energy that can raise temperature.

D Enzymes in the microorganisms work best at low temperatures.

8 · Part 1 Core Genetic modification [1]

Which action is genetic modification?

A inserting an individual gene into an organism's genetic material

B selecting two high-yield parents and breeding them together

C feeding an animal a more nutritious diet

D training an animal to respond to a signal

9 · Part 1 Core Genetic modification [1]

A crop receives a gene that makes it resistant to an insect pest. What is the intended advantage?

A the crop no longer needs fertiliser

B less crop damage from that pest

C the crop can be grown without irrigation

D the crop becomes resistant to all plant diseases

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10 · Part 1 Supplement Genetic modification [1]

Which enzyme joins a human gene into a cut bacterial plasmid?

A pectinase

B lactase

C DNA ligase

D protease

11 · Part 1 Core Biotechnology and genetic modification [4]

A company uses bacteria to produce a medically useful protein.

(a) State two biological features of bacteria that make them suitable. [2]

(b) Explain how these features help production on a large scale. [2]

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12 · Part 1 Supplement Biotechnology and genetic modification [4]

Researchers consider bacteria rather than large mammals for a genetic-modification project.

(a) Explain the usefulness of plasmids. [2]

(b) State two reasons bacterial manipulation often raises fewer ethical concerns than using large mammals. [2]

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13 · Part 1 Core Biotechnology [4]

Yeast is used in both bread-making and biofuel production.

(a) State the two main products of anaerobic respiration in yeast. [2]

(b) Explain which product is useful in each application. [2]

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14 · Part 1 Core Biotechnology [4]

Two equal masses of the same crushed fruit are treated identically except that one receives pectinase.

(a) State the expected effect of pectinase and explain why. [2]

(b) State two variables that should be kept constant. [2]

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15 · Part 1 Core Biotechnology [4]

A student tests a biological washing powder on identical protein stains.

(a) Describe two controls needed when comparing it with an enzyme-free powder. [2]

(b) Explain why a protease can improve removal of these stains. [2]

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16 · Part 1 Supplement Biotechnology [4]

Milk is treated with lactase for people who cannot digest lactose effectively.

(a) Describe the reaction catalysed by lactase. [2]

(b) Explain two advantages of using this treatment. [2]

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17 · Part 1 Supplement Biotechnology [4]

An aerobic fermenter produces mycoprotein using a fungus.

(a) Explain why oxygen and nutrient supply are controlled. [2]

(b) Explain why pH and temperature are controlled. [2]

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18 · Part 1 Supplement Genetic modification [4]

Bacteria are modified to produce a human protein.

(a) Describe how the human gene and plasmid are prepared and joined. [2]

(b) Describe what happens after recombinant plasmids are inserted into bacteria. [2]

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19 · Part 1 Supplement Genetic modification [4]

A farmer grows a herbicide-resistant crop produced by genetic modification.

(a) Explain the intended benefit during weed control. [2]

(b) Describe two possible disadvantages or concerns. [2]

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20 · Part 1 Supplement Genetic modification [4]

A rice variety is modified to improve its nutritional content.

(a) Explain one benefit for consumers and one possible benefit for farmers. [2]

(b) Describe two issues that should be considered before widespread cultivation. [2]

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